ANJALI

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About Me

AI & Machine Learning undergraduate with a passion for building intelligent solutions and solving real-world problems using data. Experienced in developing projects involving machine learning, fraud detection, recommendation systems, and AI-powered applications. Strong analytical and problem-solving skills with the ability to learn new technologies quickly and work effectively in collaborative environments.

Skills

- **Languages:** Python, SQL, Java
- **AI / ML:** NLP, Text-to-SQL Systems, LLM Applications, Anomaly Detection, Recommendation Systems
- **Frameworks:** FastAPI, Streamlit, React, Scikit-learn, TensorFlow
- **Databases & MLOps:** PostgreSQL, SQLite, Docker
- **Tools:** GitHub, VS Code
- **Soft Skills:** Analytical Thinking, Critical Thinking, Technical Communication, Research & Documentation, Teamwork, Adaptability, Time Management, Leadership

Project Experience

AI-Powered Air Quality Monitoring System (ISRO Project) — Nov 2025 – Present

- Contributing as a System Engineer and Data Analyst Intern on a team-based edge-computing air quality monitoring system developed for the Indian Space Research Organisation (ISRO).
- Developing a hybrid LSTM forecasting pipeline using physics-validated outputs to predict real-time Air Quality Index (AQI) and support regulatory compliance monitoring.
- Supporting the migration from a Physics-Informed Neural Network (PINN) to a deterministic Chemical Transport Model (CTM) using operator splitting to improve atmospheric simulation accuracy.
- Integrating ESP32-S3 edge devices via I2C and UART communication bridges to ensure reliable, non-blocking sensor data ingestion into the ML pipeline.

Argus: AI-Powered Fraud Detection System — 2026

- Developed an anomaly detection system to identify suspicious financial transactions and fraud patterns.
- Performed feature engineering and transaction behavior analysis to improve detection reliability.
- Built automated data validation pipelines to handle missing and inconsistent financial records.
- Applied drift-awareness techniques to maintain model effectiveness against evolving fraud behavior.

Reflexa-AI: Self-Reflecting AI SQL Agent — 2026

- Developed a Text-to-SQL AI agent capable of generating, executing, and self-correcting SQL queries.
- Implemented validation layers for schema mismatches, incorrect filters, and empty outputs.
- Designed a logic-based auto-correction mechanism to rewrite faulty queries when semantic errors were detected.
- Improved query reliability through reflection-based reasoning, schema awareness, and prompt refinement.

Product Recommendation System — 2025

- Developed a hybrid recommendation engine using content-based and collaborative filtering.
- Applied TF-IDF, cosine similarity, and KNN techniques to generate personalized recommendations.
- Optimized feature engineering and ranking pipelines to improve recommendation accuracy and scalability.

Certifications

Computer Graphics — NPTEL
GenAI Powered Data Analytics Job Simulation — TATA iQ, Forage

Education

Cambridge Institute of Technology — 2023 – Present
Bachelor of Engineering (Artificial Intelligence & Machine Learning) — **CGPA: 7.89**